

Average

Average: Average is defined as the sum of different data (terms) divided by total number of terms

$$\text{Average} = \frac{\text{Sum of given terms (S)}}{\text{Total number of terms (N)}}$$

Example: Find the average of given terms 2, 3, 4, 5, 6

Sol. Total number of terms = 5

$$\text{Average} = \frac{2+3+4+5+6}{5} = 4$$

Some Basic Formulae

1. Average of 'n' natural number = $\frac{(n+1)}{2}$
2. Average of 'n' even numbers = $(n+1)$
3. Average of 'n' odd numbers = n
4. Average of 'n' consecutive natural numbers = $\frac{\text{First number} + \text{Last number}}{2}$
5. Average of sum of squares of first 'n' natural numbers = $\frac{(n+1)(2n+1)}{6}$

Some Important Points:

1. If the value of each number is increased by the same value 'a', then the average of all numbers will also increase by 'a'.
2. If the value of each number is decreased by the same value 'a', then the average of all numbers will also decrease by 'a'.
3. If the value of each number is multiplied by the same value 'a', then the average of all numbers will also get multiplied by 'a'.
4. If the value of each numbers is divided by the same value 'a', then the average of all numbers will also get divided by 'a'.

Some useful Formulae

1. If the average of 'x' numbers is a and that of 'y' numbers is b, then the average of (x + y) numbers = $\frac{xa + yb}{x + y}$

Example: The average of 10 numbers is 15 and that of 15 numbers is 20. Find the average of all 25 numbers?

Sol.
$$\text{Average} = \frac{10 \times 15 + 15 \times 20}{10 + 15} = \frac{150 + 300}{25} = \frac{450}{25} = 18$$

2. If the average of 'n' quantities is equal to 'x' when a quantity is removed the average becomes 'y'. Then the value of the removed quantity is = $[n(x - y) + y]$

Example: The average age of 24 men and 1 woman is equal to 35 years. If 1 woman left, the average becomes 34 years. Find the age of woman who left the class?

Sol: Age of women = $[25(35 - 34) + 34] = 59$ years

3. If the average of marks obtained by 'n' students in an exam is 'M'. If the average marks of passed students in 'P' and that of failed students is 'F'. Then the number of students who failed in exam is $\frac{n(P - M)}{P - F}$.

$$\text{No. of failed students} = \frac{\text{Total students}(\text{Passed average} - \text{Total average})}{\text{Passed average} - \text{Failed average}}$$

Example: The average marks obtained by 125 students in an exam is 29. If the average marks of passed students is 36 and that of failed students is 11. Find the numbers of failed students?

$$\text{Sol. No of failed students} = \frac{125(36 - 29)}{36 - 11} = \frac{125 \times 7}{25} = 35$$

4. If a batsman in his n th innings makes a score of 's' and thereby increased his average by 't' then the average after 'n' innings is 's - t(n - 1)'

Example: A batsman in his 44th innings makes a score of 86 and thereby increases his average by 1. Find the average after 44 innings?

$$\text{Sol. Average after 44th innings} = (86 - 1(44 - 1)) = 86 - 43 = 43$$

Types of Questions

1. A batsman scores 34, 36, 38, 40, 42, in his five innings respectively. Find the average runs in the five innings?

$$\text{Sol. Average runs} = \frac{34 + 36 + 38 + 40 + 42}{5} = \frac{190}{5} = 38.$$

2. The average of 10 quantities is 12. The average of 6 of them is 8. What is the average of remaining four numbers.

$$\text{Sol. The required average} = \frac{10 \times 12 - 6 \times 8}{10 - 6} = 18.$$

3. Average of marks obtained by 120 candidates in a certain examination is 35. If the average marks of passed candidates is 39 and that of the failed candidates is 15, what is number of candidates who passed the examination?

Sol. Let the number of passed candidates be x .

$$\text{Then, total marks} = 120 \times 35 = 39x + (120 - x) \times 15$$

$$\text{or, } 4200 = 39x + 1800 - 15x$$

$$\text{or, } 24x = 2400$$

$$\therefore x = 100$$

$$\therefore \text{ number of passed candidates} = 100.$$

4. The average salary of the entire staff in a office is Rs. 120 per month. The average salary of officers is Rs. 460 and that of non-officers is Rs 110. If the number of officers is 15, then find the number of non-officers in the office.

Sol. Let the required number of non-officers = x

$$\text{Then, } 110x + 460 \times 15 = 120 (15 + x)$$

$$\text{or, } 120x - 110x = 460 \times 15 - 120 \times 15 = 1 (460 - 120)$$

$$\text{or, } 10x = 15 \times 340; \therefore x = 15 \times 34 = 510$$

$$\text{No. of non-officers} = \text{No. of officers} \times$$

$$\left(\frac{\text{Av. salary of officers} - \text{Mean average}}{\text{Mean average} - \text{Av. Salary of non-officers}} \right)$$

$$= 15 \left(\frac{460 - 120}{120 - 110} \right) = 510.$$

5. The average of the first and the second of three numbers is 10 more than the average of the second and the third of these numbers. What is the difference between the first and third of these three numbers?

Sol. Average of the first and the second numbers

$$= \frac{\text{First} + \text{Second}}{2}$$

Average of the second and the third numbers

$$= \frac{\text{Second} + \text{Third}}{2}$$

According to the question,

$$\frac{\text{First} + \text{second}}{2} - \frac{\text{Second} + \text{Third}}{2} = 10$$

$$\therefore \text{ First} - \text{Third} = 20$$

6. The average height of 50 students of a class is 152 cm. If 10 among them whose average height is 148 cm left the class and 10 new boys of average height 150 cm are included in the class, then what will be the new average height of the students of the class?

Sol. The total height of 50 students = $152 \times 50 = 7600$ cm
Total decrease in the height when 10 students left the class

$$= 148 \times 10 = 1480 \text{ cm}$$

Total increase in the height after 10 new students came

$$= 10 \times 150 = 1500 \text{ cm}$$

$$\text{Now, Total height of 50 students} = 7600 - 1480 + 1500 = 7620 \text{ cm}$$

$$\text{New average} = \frac{7620}{50} = 152.4 \text{ cm}$$

7. The sum of the present ages of Hari and Mohan is double the difference of their present ages. Four years ago this ratio was one and half times. Find the ratio of their ages after 12 years.

Sol. Let the present ages of Hari and Mohan be respectively x years and y years then

$$x + y = 2(x - y), \quad x = 3y \quad \dots (i)$$

$$\text{From the question, } (x - 4 + y - 4) = \frac{3}{2} \{(x - 4) - (y - 4)\}$$

$$\text{or, } 2x + 2y - 16 = 3x - 3y$$

$$\text{or, } 5y - x = 16 \quad \dots (ii)$$

Solving eqn (i) and (ii),

$$x = 24 \text{ and } y = 8$$

Hari's age after 12 years = $24 + 12 = 36$ years

Mohan's age after 12 years = $8 + 12 = 20$ years

The ratio of their ages = $36 : 20 = 9 : 5$

- 8 The present age of Geeta is 5 year more than the present age of his brother Mahesh. The present age of

their father is twice the sum of their ages. At the time of Mahesh's birth their father's age was 8 times of the age of Geeta. Find their present ages?

Sol. Let the present age of Mahesh is x years then the present age of Geeta will be $x + 5$ and the present age of their father will be $2(x + x + 5) = 4x + 10$ years.

The age of Geeta at the time of Mahesh's birth will be $x + 5 - x = 5$ years while the age of the father will be $4x + 10 - x = 3x + 10$ years

From the question,

$$3x + 10 = 5 \times 8$$

$$\text{or, } 3x = 40 - 10 = 30 \quad \text{or, } x = 10 \text{ years}$$

$$\text{and } x + 5 = 15 \text{ and } 4x + 10 = 50 \text{ years}$$

So, the ages of Mahesh, Geeta and their father are 10, 15 and 50 years respectively.

Foundation

Questions

1. A motorist covered the first 20 km of his journey in 30 minutes and the remaining 29 km in 40 minutes. His average speed in km/hr is:

(a) $41\frac{3}{4}$ (b) 42

(c) 49 (d) $57\frac{1}{6}$

2. A cyclist goes from a place to another and returns by the same route. He pedals his way uniformly with speed u while going and with speed v while returning. The average speed of his journey is :

(a) $\frac{u+v}{2}$ (b) $\frac{u-v}{2}$

(c) $\frac{1}{\frac{1}{2}\left(\frac{1}{u} + \frac{1}{v}\right)}$ (d) $\frac{1}{\frac{1}{2}\left(\frac{1}{u} - \frac{1}{v}\right)}$

3. The mean of 19 observations is 4. If one more observation of 24 is added to the data, the new mean will be:

(a) 4 (b) 5

(c) 6 (d) 7

4. A student bought 4 books for Rs. 120 from one book shop and 6 books for Rs. 150 from another. The average price (in rupees), he paid per book was :

(a) 27 (b) 27.50

(c) 135 (d) 138

5. The average of the price per kg. of rice at 10 different places was Rs. 4.85. After a week, the price per kg. was increased by 20 paise at 3 places and decreased by 10 paise at one place. The new average of price per kg is:

(a) Rs. 4.88 (b) Rs. 4.86

(c) Rs. 4.90 (d) Rs. 5.35

6. The average weight of a group of 20 boys was calculated to be 89.4 kg and it was later discovered that one weight was misread as 78 kg instead of the correct one of 87 kg. The correct average weight is :

(a) 88.95 kg (b) 89.25 kg

(c) 89.55 kg (d) 89.85 kg

7. A class has 20 boys and 30 girls. The average age of boys is 11 years and that of girls is 12 years. What is the average age of the whole class

(a) 11.6 years (b) 12 years

(c) 10 years (d) 11.2 years

8. Find the average of the following set of scores 253, 124, 255, 534, 836, 375, 101, 443, 760

(a) 427 (b) 413

(c) 141 (d) 409

9. What will be the average of even numbers between 11 to 63?

(a) 37.5 (b) 47

(c) 42 (d) 37

10. Find average of all prime numbers between 60 and 90.

(a) 72 (b) 74.7

(c) 74 (d) 73.6

11. Average age of 5 boys is 16 yr, of which that of 4 boys is 16 yr 3 months. The age of the 5th boy is

(a) 15 yr (b) 15 yr 6 months

(c) 15 yr 4 months (d) 15 yr 2 months

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12. The average age of 30 girls is 13 yr. The average of first 18 girls is 15 yr. Find out the average age of remaining 12 girls?
 (a) 12 yr (b) 10 yr
 (c) 16 yr (d) 10.5 yr
13. The average of 13 results is 60. If the average of first 7 results is 59 and that of last 7 results is 61, what will be the seventh result?
 (a) 90 (b) 50
 (c) 75 (d) 60
14. The average of nine numbers is 50. The average of the first five numbers is 54 and that of the last three numbers is 52. Then, the sixth number is?
 (a) 34 (b) 24
 (c) 44 (d) 30
15. The average age of 30 women decreases by 3 months if a new person Neha is included in place of a 25 yr old woman. Calculate the age of Neha?
 (a) 17.5 yr (b) 20 yr
 (c) 30 yr (d) 22 yr
16. The average salary per head of all workers of an institution is Rs. 60. The average salary per head of 12 officers is Rs. 400. The average salary per head of the rest is Rs. 56. Then, the total number of workers in the institution is?
 (a) 1030 (b) 1032
 (c) 1062 (d) 1060
17. A cricketer scored some runs in his continuous 9 innings. He scored 100 runs in his 10th innings and this increased his average by 8 runs. What was the average of his runs at the end of 10th innings?
 (a) 20 (b) 24
 (c) 28 (d) 32
18. The average age of three boys is 15 yr. If the ratio of their ages is 3 : 5 : 7, what is the age of the oldest boy?
 (a) 7 yr (b) 14 yr
 (c) 20 yr (d) 21 yr
19. Nine friends have a dinner in a hotel. Eight of them spent Rs. 12 each on their meals and the ninth spent Rs. 16 more than the average expenditure of all the nine. Find out the total money spent by them?
 (a) Rs. 126 (b) Rs. 135
 (c) Rs. 111 (d) Rs. 141
20. The average of certain first consecutive even number is 101. Find their sum?
 (a) 25,000 (b) 33,600
 (c) 10100 (d) 24,960
21. The average of certain first consecutive natural numbers is 20.5. Find their sum.
 (a) 282 (b) 450
 (c) 518 (d) 820
22. If the average of m numbers is n^2 and that of n numbers is m^2 , then the average of $(m + n)$ numbers is -
 (a) $\frac{m}{n}$ (b) $m + n$
 (c) mn (d) $m - n$
23. A, B, C, D & E are five consecutive even numbers. Average of A and E is 46. What is the largest number?
 (a) 52 (b) 42
 (c) 50 (d) 48
24. Of the three numbers whose average is 60, the first is one-fourth of the sum of the others. The first number is
 (a) 30 (b) 36
 (c) 42 (d) 45
25. The average salary of 15 persons is Rs. 5,500. If the salary of one person is added, the average increase to Rs. 5,700. What is the salary of this one person?
 (a) Rs. 8,700 (b) Rs. 9,500
 (c) Rs. 7,800 (d) Rs. 8,500
26. The average weight of 24 students is 35 kgs. If the weight of teacher is added the average increased by 400 gms. What is the weight of the teacher?
 (a) 45 (b) 36
 (c) 54 (d) 63
27. The average of runs of a cricket player of 10 innings was 32. How many runs must he score in his next innings so as to increase his average of runs by 4?
 (a) 76 (b) 70
 (c) 4 (d) 2
28. 10 kg of rice priced at Rs. 12 per kg is mixed with 6 kg of rice priced at Rs. 16 per kg. Find the average price of the whole mixture?
 (a) Rs. 14 per kg (b) Rs. 14.50 per kg
 (c) Rs. 13 per kg (d) Rs. 13.50 per kg
29. The average of 5 quantities is 6. The average of three of them is 4. What is the average of remaining two quantities?
 (a) 2 (b) 5
 (c) 9 (d) 7
30. The average of five numbers is 6.9. If one of the numbers is deleted, the average of the remaining numbers is 4.4. What is the value of the number deleted?
 (a) 12.5 (b) 17.2
 (c) 16.9 (d) 16.8
31. The average age of Seema, Sapna, Asha, Kavita and Atrye is 40 years. The average age of Seema and Sapna is 35 years and the average age of Asha and Kavita is 42 years. The age of Atrye is?
 (a) 48 (b) 47
 (c) 46 (d) 32
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32. The average of 11 numbers is 10.9. If the average of the first six is 10.5 and the average of last six is 11.4, find the middle (sixth) number.
 (a) 11 (b) 10.6
 (c) 11.5 (d) 10.45
33. The average temperature from Monday to Thursday is 48°C and from Tuesday to Friday, the average temperature is 52°C . If the temperature on Monday is 42°C , what was it on Friday?
 (a) 46°C (b) 58°C
 (c) 50°C (d) 45°C
34. Find the average of first 25 multiples of 5.
 (a) 65 (b) 60
 (c) 75 (d) 80
35. What was the average daily expenditure of a man in 1999 who spent Rs. 76535 in the first half year and Rs. 88165 in the last?
 (a) Rs. 450 (b) Rs. 451.32
 (c) Rs. 451.23 (d) Rs. 450.23
36. The average of 6 numbers is 8. What is the 7th number so that average becomes 10?
 (a) 22 (b) 18
 (c) 21 (d) 20
37. The average expenditure of a man for the first five months is Rs. 120 and for the next seven months is Rs. 130. His monthly average income if he saves Rs. 290 in that year, is
 (a) Rs. 160 (b) Rs. 170
 (c) Rs. 150 (d) Rs. 140
38. The average salary of 20 workers in an office is Rs. 1900 per month. If the manager's salary is added, the average becomes Rs. 2000 per month. The manager's annual salary (in Rs) is:
 (a) Rs. 24000 (b) Rs. 25200
 (c) Rs. 45600 (d) Rs. 48000
39. A club has 15 men and 25 women. The average age of men is 10 years and the average age of women is 12 year. What is the average age of the whole club?
 (a) 11.40 years (b) 11.60 years
 (c) 11.25 years (d) 12.25 years
40. The average of 12 results is 15, and the average of the first two is 14. What is the average of the rest?
 (a) 15.2 (b) 13.2
 (c) 15 (d) 16

Moderate

1. An aeroplane flies along the four sides of a square field at speeds of 200, 400, 600 and 800 km/hr. The average speed of the plane in the flight around the field in km/hr is:
 (a) 384 (b) 400
 (c) 500 (d) 284
2. The average speed of a cyclist who covers first, second and third km at 20, 16 and 12 km/hr respectively (in km/hr) is:
 (a) 16.24 (b) 16
 (c) 15.66 (d) 15.32
3. In a class of 100 students, the mean marks obtained in a subject is 30 and in another class of 50 students the mean marks obtained in the same subject is 60. The mean marks obtained by the students of two classes taken together is :
 (a) 40 (b) 45
 (c) 48 (d) 50
4. The mean weight of 34 students of a school is 42 kg. If the weight of the teacher be included, the mean rises by 400 g. Find the weight (in kg) of the teacher?
 (a) 66 (b) 56
 (c) 55 (d) 57
5. The average weight of 21 boys was recorded as 64 kg. If the weight of the 'teacher' was added, the average increased by one kg. What was the teacher's weight?
 (a) 86 kg (b) 64 kg
 (c) 72 kg (d) 98 kg
6. Roshan bought 5 pants at Rs. 25 each, 10 shirts at Rs. 50 each and 15 ties at Rs. 35 each. Find the average price of all the articles.
 (a) Rs. 38.33 (b) Rs. 45
 (c) Rs. 60 (d) Rs. 45.33
7. The mean weight of 150 students in a class is 60 kg. The mean weight of boys is 70 kg and that of girls is 55 kg. What is the number of boys in the class?
 (a) 50 (b) 60
 (c) 75 (d) 100
8. The average of the test scores of a class of 'm' students is 70 and that of 'n' students is 91. When the scores of both the classes are combined, the average is 80. What is n/m ?
 (a) $\frac{11}{10}$ (b) $\frac{13}{10}$
 (c) $\frac{10}{13}$ (d) $\frac{10}{11}$

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9. The average marks of 120 students are 35. If average of passed students was 39 and failed students was 15, then Find the number of students who have passed?
- (a) 80 (b) 100
(c) 120 (d) 140
10. A person covers 9 km with a speed of 3 km/h, 25 km with a speed of 5 km/h and 30 km with a speed of 10 km/h. Find out the average speed of person?
- (a) $5\frac{9}{11}$ km / h (b) $11\frac{5}{9}$ km / h
(c) $9\frac{5}{11}$ km / h (d) $5\frac{5}{11}$ km / h
11. Five years ago, the average age of P and Q was 15 yr. Now, average age of P, Q and R is 20 yr. What would be the age of R after 10 yr?
- (a) 35 yr (b) 40 yr
(c) 30 yr (d) 50 yr
12. The average salary of the entire staff in an office is Rs. 500 per day. The average salary of officers is Rs. 750 and that of non-officers is Rs. 250. If the number of officers is 15, then find the number of non-officers in the office?
- (a) 25 (b) 10
(c) 15 (d) 20
13. Three year ago, the average age of a family of five members was 27 yr. The present average age is also 27 yr after the inclusion of a child into the family. Find the age of the child?
- (a) 16 yr (b) 12 yr
(c) 24 yr (d) 20 yr
14. The average age of 5 sisters is 20 yr. If the age of the youngest sister be 4 yr, what was the average age of the group of sisters at the birth of the youngest sister?
- (a) 25 yr (b) 15 yr
(c) 18 yr (d) 20 yr
15. The average weight of 4 men A, B, C and D, is 67 kg. The 5th man E is included and the average weight decreases by 2 kg. A is replaced by F. The weight of F is 4 kg more than E. Average weight decreases because of the replacement of A and now the average weight is 64 kg. Find the weight of A?
- (a) 78 kg (b) 66 kg
(c) 75 kg (d) 58 kg
16. Average of a, b and c is 11; average of c, d and e is 17; average of e and f is 22 and average of e and c is 17. Find out the average of a, b, c, d, e and f?
- (a) $15\frac{2}{3}$ (b) $18\frac{1}{2}$
(c) $21\frac{1}{3}$ (d) $16\frac{1}{2}$
17. The average age of a man and his two twin born on the same day is 30 yr. The ratio of the ages of father and one son is 5 : 2, what is the father's age?
- (a) 50 yr (b) 30 yr
(c) 45 yr (d) 20 yr
18. If the average of the ages of Rakesh and Mohan is 15, average of the ages of Mohan and Ramesh is 12 and the average of the ages of Rakesh and Ramesh is 13, then the age of Mohan, is?
- (a) 16 (b) 13
(c) 14 (d) 12
19. The average of three numbers is 28. If first number is half of the second and the 3rd number is twice the second, then the third number is?
- (a) 48 (b) 36
(c) 24 (d) 18
20. The average temperature from Monday to Wednesday is 37°C while the average temperature from Tuesday to Thursday is 34°C . The temperature of Thursday is $\frac{4}{5}$ times to that of Monday. Find the temperature of Thursday?
- (a) 36°C (b) 33°C
(c) 37°C (d) 34°C
21. There are 6 consecutive odd numbers in increasing order. The difference between the average of squares of the last four numbers and the first four numbers is 64. Then the average of all the six numbers is?
- (a) 8 (b) 7
(c) 6 (d) 5
22. The average of marks obtained by 150 candidates in a certain examination is 50. If the average marks of passed candidates is 55 and that of the failed candidates is 25, what is the number of candidates who passed the examination?
- (a) 125 (b) 120
(c) 140 (d) 160
23. The average age of 24 boys and their teacher is 15 years. When the teacher's age is excluded, the average age decreases by 1 year. The age of the teacher is?
- (a) 38 years (b) 39 years
(c) 40 years (d) 41 years
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24. The average age of 8 members of a group is 40 years. The age of 1 member is 55 yrs. After his retirement a new member comes in his place of age 39. What is the effect of his retirement on the average age of group?
- (a) 2 (b) 0
(c) 3 (d) 5
25. The average age of 11 players of a cricket team is increased by 2 months when two of them aged 18 years and 20 years are replaced by two new players. The average age of the new players is?
- (a) 19 years 1 month (b) 19 years 6 months
(c) 19 years 11 months (d) 19 years 5 months
26. In a school, the average age of students is 6 years, and the average age of 12 teachers is 40 years. If the average age of the combined group of all the teachers and students is 7 years, then the number of students is?
- (a) 396 (b) 400
(c) 408 (d) 416
27. A batsman has a certain average in 11 innings. In the 12th innings, he scores 90 runs and his average decreases by 5. After the 12th innings what is his average?
- (a) 145 (b) 130
(c) 135 (d) 140
28. A cricketer gives 12.4 runs per wicket. He gives 26 runs and takes 5 wicket in a match after which his average becomes 12 per wicket. How many wickets had been taken till the last match?
- (a) 85 (b) 90
(c) 65 (d) 72
29. The average of a batsman after 25 innings was 56 runs per innings. If after the 26th inning his average increased by 2 runs, then find his score in the 26th inning?
- (a) 100 (b) 102
(c) 108 (d) 98
30. The average age of a class of 30 students and a teacher reduce by 0.5 years if we exclude the teacher. If the initial average is 14 years, find the age of the class teacher?
- (a) 29 years (b) 30 years
(c) 31 years (d) 32 years
31. The average marks of a group of 20 students on a test is reduced by 4 when the topper who scored 90 marks is replaced by a new students. Find the marks of the new student.
- (a) 10 (b) 51
(c) 56 (d) 86
32. The average age of x and y is 20 years. If z were to replace x, the average would be 19 and if z were to replace y, the average would be 21. Find the ages of x, y and z.
- (a) 22, 18, 20 (b) 18, 22, 20
(c) 20, 18, 20 (d) 18, 20, 22
33. The average age of a family of 6 members is 22 years. If the age of the youngest member be 7 years, then what was the average age of the family at the birth of the youngest member?
- (a) 18 (b) 15
(c) 13 (d) 26
34. The average age of 10 men is increased by 2 years when one of them of 25 years age, is replaced by new man. What is the age of new man?
- (a) 50 (b) 45
(c) 35 (d) 25
35. The mean of the marks obtained by 100 students is 60. If the marks obtained by one of the students was incorrectly calculated as 75, whereas the actual marks obtained by him were 65, what is the correct mean of the marks obtained by the students?
- (a) 59.9 (b) 59
(c) 58.5 (d) 59.4
36. A, B, C and D are four consecutive even numbers respectively and their average is 65. What is the product of A and D?
- (a) 3968 (b) 4092
(c) 4216 (d) 4352
37. The average weight of 36 students is 50 kg. If was found later that the weight of one of the student was misread as 73 kg. whereas his actual weight was 37 kg. Find the correct average.
- (a) 47 (b) 49.5
(c) 49 (d) 51
38. The average score of Sachin after 25 innings is 46 runs per innings. If after the 26th innings, his average runs increased by 2 runs, then what is his score in the 26th innings?
- (a) 48 (b) 84
(c) 98 (d) 89
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39. 3 years ago the average age of a family of 5 members was 17 years, with the birth of a new baby, the average age of six members remains the same even today. Find the age of the new baby.
- (a) 2 years (b) 3 years
(c) 5 years (d) can't be determined
40. The average marks of a student in four subjects is 75. If the student obtained 80 marks in the 5th subject then the new average is?
- (a) 80 (b) 76
(c) 92 (d) 95

Difficult

1. The average weight of A, B and C is 84 kgs. If D joins, the average weight now is 80 kgs. If another person E who is 3 kgs heavier than D replaces A then the average weight of B, C, D and E becomes 79 kgs. what is the weight of A?
- (a) 70 kg (b) 75 kg
(c) 65 kg (d) 55 kg
2. A batsman has an average of 30 runs in his 42 inning. The difference between his maximum and minimum score is 100. If these two innings are removed his average for 40 innings comes down to 28. What is his minimum score?
- (a) 20 (b) 30
(c) 45 (d) 40
3. Five years ago the average age of husband and wife was 23 years. Today the average age of husband, wife and child is 20 yrs. How old is child?
- (a) 3 yrs (b) 4 yrs
(c) 12 yrs (d) 2 yrs
4. The average age of a husband and wife, who were married 7 years ago, was 25 years at the time of their marriage, Now, the average age of the family, including husband, wife and a child, born during the interval, is 22 years. What is the present age of the child?
- (a) 4 yrs. (b) 2 yrs.
(c) 3 yrs. (d) 5 yrs.
5. The average age of mother, father and son was 42 yrs. at the time of the marriage of the son. After one year an infant was born in the family and after 6 yrs of marriage the average age of the family is 36 years. What was the age of the bride at the time of marriage.
- (a) 25 yrs. (b) 23 yrs.
(c) 22 yrs. (d) 24 yrs.
6. There were 35 students in a hostel. If the number of students increases by 7, the expenses of the mess increase by Rs. 42 per day while the average expenditure per head decreases by Rs. 1. Find the original expenditure of the mess?
- (a) Rs. 420 (b) Rs. 520
(c) Rs. 450 (d) Rs. 550
7. Ashish has earned an average of 4200 dollars for the first eleven months of the year. If he justifies his staying on in the US on the basis of his ability to earn at least 5000 dollars per month for the entire year, how much should he earn (in dollars) in the last month to achieve his required average for the whole year?
- (a) 14,600 (b) 5,800
(c) 12,800 (d) 13800
8. The sum of three numbers is 98. If the ratio between the first and second is 2 : 3 and that between the second and the third is 5 : 8, then the second number is
- (a) 30 (b) 20
(c) 58 (d) 48
9. The average salary per head of all the workers in a company is Rs. 95. The average salary of 15 officers is Rs. 525 and the average salary per head of the rest is Rs. 85. Find the total number of workers in the workshop.
- (a) 660 (b) 580
(c) 650 (d) 460
10. A batsman makes a score of 270 runs in the 87th inning and thus increases his average by a certain number of runs that is a whole number. Find the possible values of the new average.
- (a) 98 (b) 184
(c) 12 (d) All of these
11. 19 persons went to a hotel for a combined dinner party. 13 of them spent Rs. 79 each on their dinner and the rest spent Rs. 4 more than the average expenditure of all the 19. What was the total money spent by them.
- (a) 1628.4 (b) 1536.07
(c) 1492 (d) 1496
12. The average age of a group of persons going for a picnic is 16.75 years. 20 new persons with an average age of 13.25 years join the group on the spot due to which the average of the group becomes 15 years. Find the number of persons initially going for the picnic?

-
- (a) 24 (b) 20
(c) 15 (d) 18
13. The average salary of workers in AMS careers is Rs. 2,000, the average salary of faculty being Rs. 4,000 and the management trainees being Rs. 1,250. The total number of workers could be
(a) 450 (b) 300
(c) 110 (d) 500
14. 10 years ago the average age of all the 25 teachers of the Girls college was 45 years. 4 years ago, the principal has retired from her post at the age of 60 year. So after one year a new principal whose age was 54 years recruited from outside. The present average age of all the teachers is, if principal is also considered as a teacher:
(a) $54\frac{18}{25}$ years (b) $55\frac{17}{25}$ years
(c) $49\frac{1}{2}$ years (d) $49\frac{2}{3}$ years
15. The average weight of all the 11 players of Indian cricket team is 50 kg. If the average of first six lightest weight players is 49 kg and that of the six heaviest players is 52 kg. The average weight of the player which lies in the sixth position in the list of players when all the 11 players are arranged in the order of increasing or decreasing weights:
(a) 56 kg (b) 52 kg
(c) 51 kg (d) 50 kg
16. The average age of Sachin and Ganguli is 35 years. If Kaif replaces Sachin, the average age becomes 32 years and if Kaif replaces Ganguli, then the average age becomes 38 years. If the average age of Dhoni and Irfan be half of the average age of Sachin, Ganguli and Kaif, then the average age of all the five people is:
(a) 28 years (b) 32 years
(c) 25 years (d) 26 years
17. The average age of a family of 6 members 4 year ago was 25 years. Meanwhile a child was born in this family and still the average age of the whole family is same today. The present age of the child is:
(a) 2 years (b) $1\frac{1}{2}$ years
(c) 1 year (d) data insufficient
18. Amitabh's average expenditure for the January to June is Rs. 4200 and he spends Rs. 1200 in January and Rs. 1500 in July. The average expenditure for the month of February to July is:
(a) Rs. 4250 (b) Rs. 4520
(c) Rs. 4060 (d) Rs. 4200
19. The average price of 80 computers in an electronic shop is Rs. 30,000. If the highest and lowest price computers are sold out then the average price of the remaining 78 computers is Rs. 29,500. The cost of the highest price computer is Rs. 80,000. The cost of lowest price computer is:
(a) Rs. 19,000 (b) Rs. 20,000
(c) Rs. 29,000 (d) can't be determined
20. Eleven years earlier the average age of a family of 4 members was 28 years. Now the age of the same family with six members is yet the same, even when 2 children were born in this period. If they belong to the same parents and the age of the first child at the time of the birth of the younger child was same as there were total family members just after the birth of the youngest member of this family, then the present age of the youngest member of the family is :
(a) 3 years (b) 5 years
(c) 6 years (d) 4 years
21. Mr. Patel walked 6 km to reach the station from his house, then he boarded a train whose average speed was 60 km/hr and thus he reached his destination. In this way he took total of 3 hours. If the average speed of the entire journey was 32 km/hr then the average speed of walking is :
(a) 3 km/hr (b) 4.5 km/hr
(c) 4 km/hr (d) 5 km/hr
22. In a set of prime and composite numbers, the composite numbers are twice the number of prime number and the average of all the numbers of the set is 9. If the number of prime numbers and composite numbers are exchanged then the average of the set of numbers is increased by 2. If during the exchange of the numbers the average of the prime numbers and composite numbers individually remained constant, then the ratio of the average of composite numbers to the average of prime numbers (initially) was:
(a) $\frac{7}{13}$ (b) $\frac{13}{7}$
(c) $\frac{9}{11}$ (d) $\frac{7}{11}$
-

23. The average marks of Sameer decreased by 1, when he replaced the subject in which he has scored 40 marks by the other two subjects in which he has just scored 23 and 25 marks respectively. later he has also included 57 marks of Computer Science, then the average marks increased by 2. How many subjects were there initially?
- (a) 6 (b) 12
(c) 15 (d) can't be determined
24. The average age of Donald, his wife and their two children is 23 years. His wife is just 4 year younger than Donald himself and his wife was 24 years old when his daughter was born. He was 32 years old when his son was born. Th average age of Donald and his daughter is :
- (a) 25 years (b) 22.5 years
(c) 26 years (d) can't be determined
25. There are 6 consecutive odd numbers in increasing order. The difference between the average of the squares of the first 4 numbers and the last four numbers is 64. If the sum of the squares of the first and the last element (i.e., odd numbers) is 178, then the average of all the six numbers is:
- (a) 7 (b) 8
(c) 9 (d) 10
26. The average age of board of directors of a company, having 10 directors was 48 years. Coincidentally when a director aged 53 resigned from the board of directors, another director died on the same day. So a new director joined the board of directors aged 34. Next year in the same month the average age of all the 9 directors was found to be 46 years. The age of the late (i.e., dead) director at the time of his death was :
- (a) 56 years (b) 53 years
(c) 57 years (d) 61 years

Previous Year (Memory Based)

1. In a cricket team, the average age of 11 players and the coach is 18 yr. If the age of the coach is not considered, then the average decreases by 1 yr. Find out the age of the coach?
- (a) 29 yr (b) 21 yr
(c) 25 yr (d) 30 yr
2. The average salary of the entire staff in an office is Rs 200 per day. The average salary of officers is Rs 550 and that of non-officers is Rs 120. If the number of officers is 16, then find the numbers of non-officers in the office?
- (a) 70 (b) 30
(c) 50 (d) 40
3. A person covers 20 km distance with a speed of 5 km/h, then he covers the next 15 km with a speed of 3 km/h and the last 10 km is covered by him with a speed of 2 km/h. Find out his average speed for the whole journey.
- (a) $3\frac{3}{14}$ km / h (b) $3\frac{2}{14}$ km / h
(c) $1\frac{3}{10}$ km / h (d) $1\frac{3}{12}$ km / h
4. Mr. Sharma travels by car and covers 25% of his journey with a speed of 10 km/h, 45% of his journey with a speed of 5 km/h and remaining 30% of his journey with a speed of 15 km/h. What will be the average speed of Mr. Sharma for the whole journey?
- (a) 5.40 km/h (b) 7.40 km/h
(c) 2.40 km/h (d) 8.40 km/h
5. The average runs scored by a cricketer in 42 innings, is 30. The difference between his maximum and minimum scores in an innings is 100. If these two innings are not taken into consideration, then the average score of remaining 40 innings is 28. Calculate the maximum runs scored by him in an innings?
- (a) 125 (b) 120
(c) 110 (d) 100
6. A class is divided into two sections A and B. Passing average of 20 students of section A is 80% and passing average of 30 students of section B is 70%. What is the passing average of both the sections?
- (a) 72% (b) 74%
(c) 75% (d) 77%
7. The arithmetic mean of the scores of a group of students in a test was 52. The brightest 20% of them secured a mean score of 80 and the dullest 25%, a mean score of 31. The mean score of remaining 55% is
- (a) 54.6% (b) 45%
(c) 50% (d) 51.4%
8. A person bought some oranges worth Rs 36 from each of the five markets at Rs 1, Rs 1.50, Rs 1.80, Rs 2 and Rs 2.25 per orange, respectively. What is the average price of an orange?
- (a) Rs 1.91 (b) Rs 2.00
(c) Rs 1.58 (d) Rs 1.80

9. The average weight of the students in four sections A, B, C and D is 60 kg. The average weight of the students of A, B, C and D individually are 45 kg, 50 kg, 72 kg and 80 kg, respectively. If the average weight of the students of sections A and B together is 48 kg and that of B and C together is 60 kg, what is the ratio of the number of students in sections A and D?
(a) 12 : 7 (b) 4 : 3
(c) 3 : 2 (d) 8 : 5
10. The average age of students of a class is 15.8 yr. The average age of boys in the class is 16.4 yr and that of the girls is 15.4 yr. Find out the ratio of the number of boys to the number of girls in the class.
(a) 3 : 1 (b) 5 : 2
(c) 2 : 3 (d) 3 : 7
11. What will be the average of first five prime numbers?
(a) 5.6 (b) 7.1
(c) 7.6 (d) 4.6
12. A cricket player scored 53, 52, 11, 0, 5 not out, 104 not out and 35 runs in 7 innings of a series. What is the average of run scored by player in that series?
(a) 62 (b) 52
(c) 70 (d) 80
13. A joint family consists of 5 males, 4 females and some children. The average consumption per person per month is 1.5 kg. If the average consumption for male, female and children is respectively 1 kg, 1.25 kg and 2 kg then what is the number of children in the family?
(a) 9 (b) 5
(c) 7 (d) 8
14. The average temperature of Monday, Tuesday and Wednesday was 32°C. The average temperature of Tuesday, Wednesday and Thursday was 34°C. If the temperature of Monday be 29°C. Find the temperature on Thursday?
(a) 37° (b) 26°
(c) 25° (d) 35°
15. The average of 40 observations was 28. It was later found that in two observations, 42 was taken instead of 24 and 12 instead of 62. What is the correct average?
(a) 28.5 (b) 28.8
(c) 26.8 (d) 23.5
16. The average of 15 numbers is 27 in which the average of first 7 is 25 and the average of last seven is 31. Find the 8th number?
(a) 15 (b) 11
(c) 13 (d) 14
17. The average age of 9 boys is 15 years and when the teacher's age is included the average age becomes 17. What is the age of the teacher?
(a) 35 years (b) 16 years
(c) 40 years (d) 15 years
18. The average height of 50 students of a class is 152 cm. If 10 among them whose average height is 148 cm left the class and 10 new boys of average height 150 cm are included in the class, then what will be the new average height of the students of the class?
(a) 162.4 cm (b) 112.5 cm
(c) 152.4 cm (d) 122.6 cm
19. The average age of 5 boys is 12 years. The average age of other 3 boys is 16 years. What will be the average of all 8 boys?
(a) $13\frac{1}{2}$ years (b) $12\frac{1}{2}$ years
(c) 14 years (d) 13 years
20. The average income of 40 persons is Rs. 4200 and the average income of other 35 persons is Rs. 4,000. What is the average income of the entire group (in Rupees)?
(a) 4100 (b) $4106\frac{1}{3}$
(c) $4106\frac{2}{3}$ (d) $4108\frac{1}{3}$
21. In a T.V. factory an average of 60 pieces of T.V. is produced in first 25 days of any month. Due to the illness of few workers the average production per day lower downs to 58 for whole month. What is the average production of last 5 days?
(a) $59\frac{2}{3}$ (b) 60
(c) 48 (d) 53
22. The average annual income (in Rupees) of some agriculture labours is S and the average annual income of some other labours is T. The number of agriculture labours is 11 times more than that of other labours. Then what will be the average monthly income (in Rupees) of all the labours?
(a) $\frac{S+11T}{12}$ (b) $\frac{S+T}{2}$
(c) $\frac{11S+T}{12}$ (d) $\frac{11S+T}{144}$
23. The average consumption of rice per person per month in a family of 8 adults and some kids is 10.8 kg, where the average consumption per person for adult is 15 kg and for kids is 6 kg. What is the number of kids in the family?
(a) 8 (b) 6
(c) 7 (d) 9
24. The average salary of all the workers in a workshop in Rs. 8000. Among them the average salary of 7 technicians is Rs. 12000 and the average salary of rest labours is Rs. 6000. What is the number of all the workers in the workshop?

-
- | | | | |
|--------|--------|---------|---------|
| (a) 14 | (b) 21 | (a) 296 | (b) 320 |
| (c) 23 | (d) 22 | (c) 324 | (d) 285 |
25. The average marks achieved by 120 aspirants in any exam is 35. The average marks achieved by passed student is 39 and the average marks achieved by failed students is 15. What is the number of passed student in the exam ?
- | | |
|---------|---------|
| (a) 100 | (b) 120 |
| (c) 150 | (d) 140 |
26. The average wages of 500 workers is Rs. 200. But it was later found that for two workers the wages was wrongly put as Rs. 180 and Rs 20 instead of Rs. 80 and Rs. 220 respectively. Then what is the correct average wages?
- | | |
|----------------|----------------|
| (a) Rs. 200.10 | (b) Rs. 200.20 |
| (c) Rs. 200.50 | (d) Rs. 201.00 |
27. In any year the monthly average income of a person is Rs. 3400. The average income of first eight month is Rs. 3160 and the average income of last 5 months is Rs. 4120. What is the income in 8th month of the year?
- | | |
|---------------|--------------|
| (a) Rs. 5080 | (b) Rs. 6080 |
| (c) Rs. 25180 | (d) Rs. 3880 |
28. A cricket player whose bowling average was 24.85 runs per wicket, takes 5 wicket for 52 runs in a match. Due to this his average decreases by 0.85. What will be the number of wickets taken by him till the last match?
- | | |
|--------|--------|
| (a) 64 | (b) 72 |
| (c) 85 | (d) 96 |
29. A batsman has a certain average of runs for his 19 innings. In 20th innings he makes 100 runs and his average increases by 2 runs then what is the average of 20 innings?
- | | |
|--------|--------|
| (a) 60 | (b) 61 |
| (c) 62 | (d) 64 |
30. In a city the rainfall recorded in inch for five consecutive years was 20.54 inch, 33.10 inch, 11.62 inch, 19.20 inch and 21.74 inch. What was the average rainfall ?
- | | |
|----------------|----------------|
| (a) 21.44 inch | (b) 21.24 inch |
| (c) 20.24 inch | (d) 17.94 inch |
31. The difference between the average of three consecutive odd number and 2 consecutive even numbers thereafter is 4. What is the first even number?
- | | |
|--------|---------------------------|
| (a) 10 | (b) 12 |
| (c) 14 | (d) Can not be determined |
32. The sum of 8 consecutive numbers of set -A is 376. What is the sum of 5 consecutive numbers of another set whose smallest number is 15 more than that of the average of set-A ?
- | | |
|---------|---------|
| (a) 296 | (b) 320 |
| (c) 324 | (d) 285 |
33. There were 35 students in hotel. If the number of students increases by 7 then the expense of fooding increases by Rs. 42 per day while average expenditure of the boys decreases by Rs. 1. What was the initial expenditure on fooding per day ?
- | | |
|-------------|-------------|
| (a) Rs. 400 | (b) Rs. 442 |
| (c) Rs. 432 | (d) Rs. 420 |
34. The average of first 5 natural numbers is m. If next 3 natural numbers are included, then the average of these 8 number will how much more than m ?
- | | |
|---------|---------|
| (a) 2 | (b) 1 |
| (c) 1.4 | (d) 1.5 |
35. If the average of m numbers is n^2 and n numbers is m^2 then the average of (m + n) numbers will be.
- | | |
|-------------------|-----------|
| (a) $\frac{m}{n}$ | (b) m + n |
| (c) mn | (d) m - n |
36. There are 50 students in a class. One boy among them, whose weight is 51 kg leaves the class and a new boy admits in the class. Due to this the average weight of the class increases by $\frac{1}{2}$ kg. The weight of newly admitted student is?
- | | |
|-----------|-----------|
| (a) 73 kg | (b) 76 kg |
| (c) 74 kg | (d) 75 kg |
37. The average marks of a student in the first three test is 'N'. If in the fourth test he gets 20 marks more than that of the average of previous tests then what is the average of first four test?
- | | |
|------------|------------|
| (a) N + 20 | (b) N + 10 |
| (c) N + 4 | (d) N + 5 |
38. A student finds the average of ten number of two digits. However he wrongly copies one number by exchaning its digits. Therefore he gets the answer 1.8 less than the correct answer. What will be the difference of the digit he wrongly put?
- | | |
|-------|-------|
| (a) 2 | (b) 3 |
| (c) 4 | (d) 5 |
39. The average of 8 continuous numbers is 6.5. What is the average of the largest and the smallest number?
- | | |
|---------|---------|
| (a) 4 | (b) 6.5 |
| (c) 7.5 | (d) 9 |
40. The average of 25 observations is recorded as 13. Later it was found that one observation with value 73 was entered wrongly as 48. The new average is
- | | |
|----------|----------|
| (a) 12.6 | (b) 14 |
| (c) 15 | (d) 13.8 |
-

Foundation

Solutions

1. (b); Total distance covered = $(20 + 29)$ km = 49 km
Total time taken = $(30 + 40)$ min = 70 min
Average speed = $\left(\frac{49}{70} \times 60\right) = 42$ km/h
2. (c); Let one way distance be 'x'.
 $\therefore$ Total distance = $2x$
Total time taken = $\left(\frac{x}{u} + \frac{x}{v}\right)$
 $\therefore$ Average speed = $\frac{2x}{\frac{x}{u} + \frac{x}{v}} = \frac{1}{\frac{1}{2}\left(\frac{1}{u} + \frac{1}{v}\right)}$
3. (b); New mean = $\left(\frac{19 \times 4 + 24}{20}\right) = \frac{100}{20} = 5$
4. (a); Cost of 10 books = Rs $(120 + 150)$ = Rs 270
Average cost = Rs. $\left(\frac{270}{10}\right)$ = Rs. 27
5. (c); The sum of price of per kg Rice at 10 different places = $4.85 \times 10 = 48.5$
Increased price = $20 \times 3 - 10 \times 1 = 50$ paise
 $\therefore$ Sum of the increased price of rice at 10 different places = $48.50 + 0.50 = 49.0$ Rs.
New Average = $\frac{49}{10} = 4.90$ Rs.
6. (d); Total weight of 20 boys = $89.4 \times 20 = 1788$ kg
Increase in weight due to misreading = $87 - 78 = 9$ kg
Correct weight of 20 boys = $1788 + 9 = 1797$ kg
Correct average weight = $\frac{1797}{20} = 89.85$ kg
Shortcut:
Increase in weight = 9 kg.
New average = $89.4 + \frac{9}{20} = 89.85$ kg
7. (a); Sum of the Age of 20 boys = $20 \times 11 = 220$ years
Sum of the Age of 30 girls = $30 \times 12 = 360$ years
Total age of 20 boys and 30 girls = $220 + 360 = 580$ years.
Average age of the whole class = $\frac{580}{50} = 11.6$ years
8. (d); Average
$$= \frac{253 + 124 + 255 + 534 + 836 + 375 + 101 + 443 + 760}{9}$$

$$= 409$$
9. (d); Total even no = $\frac{63 - 11}{2} = \frac{52}{2} = 26$
Sum of the even number between is 11 to 63
$$= \frac{26}{2} [2 \times 12 + (26 - 1) \times 2] = 13 [24 + 50]$$

$$= 13 \times 74 = 962$$

Average of even numbers between 11 to 63
$$= \frac{962}{26} = 37$$
10. (b); Average of all prime numbers between 60 to 90
$$= \frac{61 + 67 + 71 + 73 + 79 + 83 + 89}{7} = \frac{523}{7} = 74.7$$
11. (a); Total age of 5 boys = $16 \times 5 = 80$ years
Total age of 4 boys = $\left(16 \frac{3}{12}\right) \times 4 = 16.25 \times 4 = 65$
The age of the 5th boy = $80 - 65 = 15$ years
12. (b); Total age of 30 girls = $30 \times 13 = 390$ years
Total age of 18 girls = $18 \times 15 = 270$ years
 $\therefore$ Average Age of remaining 12 girls = $\frac{390 - 270}{12}$
$$= \frac{120}{12} = 10$$
 years.
13. (d); Sum of the first 13 Results = $13 \times 60 = 780$
Sum of the first 7 Results = $59 \times 7 = 413$
Sum of the last 7 Results = $7 \times 61 = 427$
 $\therefore$ 7th Result = Sum of first 7 Results + sum of last 7 Results - sum of 13 Results = $413 + 427 - 780 = 840 - 780 = 60$
14. (b); The sum of 9 numbers = $9 \times 50 = 450$
Sixth number = sum of 9 numbers - (Sum of first 5 numbers + sum of last 3 numbers)
$$= 450 - (54 \times 5 + 52 \times 3) = 450 - (270 + 156)$$

$$= 450 - 426 = 24$$
15. (a); Average decreases by 3 months or $\frac{3}{12}$ years.
Sum of the age decreased = $30 \times \frac{3}{12} = \frac{15}{2}$
$$= 7.5$$
 years
 $\therefore$ Age of Neha = $25 - 7.5 = 17.5$

16. (b); Let the total number of workers = x

According to the question

$$60x = 12 \times 400 + 56(x - 12)$$

$$60x - 56x = 4800 - 672$$

$$4x = 4128 \Rightarrow x = 1032$$

17. (c); Let average of 9 innings = x, Total run = 9x

According to the question

$$\frac{9x + 100}{10} = x + 8$$

$$9x + 100 = 10x + 80 \Rightarrow x = 20$$

$$\text{Average after 10 innings} = 20 + 8 = 28$$

Shortcut:

Average at the end of 10th innings

$$= (100 - 9 \times 8) = 28 \text{ runs.}$$

18. (d); Total age of all the three boys = $3 \times 15 = 45$ years

$$\begin{aligned} \text{Given ratio of ages} &= 3 : 5 : 7 = \frac{7}{3+5+7} \times 45 \\ &= 21 \text{ years.} \end{aligned}$$

19. (a); Let average expenditure of 9 person = x

$\therefore$ According to the question

$$12 \times 8 + (x + 16) = 9x \Rightarrow 8x = 112$$

$$\Rightarrow x = \frac{112}{8} = 14$$

$$\text{Total money spent} = 9x = 9 \times 14 = 126 \text{ Rs.}$$

20. (c); Let the total consecutive even number = x

$\therefore$ Average of consecutive even no = $x + 1$

$$x + 1 = 101 \Rightarrow x = 100$$

$$\therefore \text{Sum} = 101 \times 100 = 10100$$

21. (d); Let the total consecutive natural number = N

$\therefore$ Average of consecutive natural number

$$= \frac{N+1}{2} \Rightarrow \frac{N+1}{2} = 20.5 \Rightarrow N+1 = 41 \Rightarrow N = 40$$

$$\therefore \text{Sum} = 40 \times 20.5 = 820$$

22. (c); Sum of m numbers = $n^2 \times m$

$$\text{Sum of n numbers} = m^2 \times n$$

$$\text{Average of } (m+n) = \frac{n^2 \times m + m^2 \times n}{m+n}$$

$$= \frac{mn(n+m)}{(n+m)} = mn$$

23. (c); Let A is x

$\therefore$ Then E = x + 8

According to the question

$$\frac{x+x+8}{2} = 46 \Rightarrow 2x = 84 \Rightarrow x = 42$$

$$\text{Largest number is} = 42 + 8 \Rightarrow 50$$

24. (b); Let the First no = x

Sum of the other numbers = 4x

According to the question

$$\frac{x+4x}{3} = 60 \Rightarrow 5x = 180 \Rightarrow x = 36$$

25. (a); Sum of the Salary of 15 person is = 5500×15
= 82500

$$\text{Sum of the Salary of 16 person is} = 5700 \times 16 = 91200$$

$$\text{Salary of this one person is} = 91200 - 82500 = 8700$$

26. (a); Total weight of 24 students = $24 \times 35 = 840$ kg
Total weight after teacher is added = 25×35.4
= 885 kg

$$\text{Weight of teacher} = 885 - 840 = 45 \text{ kg}$$

27. (a); Let the run scored in 11th innings = x

$\therefore$ According to the question

$$10 \times 32 + x = 11 \times (32 + 4)$$

$$320 + x = 36 \times 11 \Rightarrow x = 396 - 320 = 76$$

Shortcut:

He must score 32 runs and some extra runs.

So, in the next innings he should score:

$$32 + 11 \times 4 = 76$$

28. (d); Total price of 10 kg Rice = $12 \times 10 \Rightarrow 120$

$$\text{Total price of 6 kg Rice} = 16 \times 6 \Rightarrow 96$$

$$\therefore \text{Average} = \frac{120+96}{16} = \frac{216}{16} = 13.50 \text{ kg}$$

29. (c); Let the average of remaining two quantities is y.

According to the question

$$\frac{3 \times 4 + 2 \times y}{5} = 6 \Rightarrow 12 + 2y = 30 \Rightarrow y = 9$$

Hence the average of remaining two quantities is 9.

30. (c); Average of five numbers

$$\Rightarrow \frac{a+b+c+d+e}{5} = 6.9 \Rightarrow a+b+c+d+e = 34.5$$

Let the number is deleted be e. The average of remaining 4 number is

$$\Rightarrow \frac{a+b+c+d}{4} = 4.4 \Rightarrow a+b+c+d = 17.6$$

$$\text{Value of deleted number} = 34.5 - 17.6 = 16.9$$

31. (c); Total age of Seema + Sapna + Asha + Kavita + Atrye = $40 \times 5 = 200$

$$\text{Seema + Sapna} = 35 \times 2 = 70$$

$$\text{Asha + Kavita} = 42 \times 2 = 84$$

$$\therefore \text{Age of Atrye} = 200 - (70 + 84) = 46 \text{ years}$$

32. (c); Total first 6 number = $10.5 \times 6 = 63$
 Total of last 6 number = $11.4 \times 6 = 68.4$
 Total of all 11 number = $11 \times 10.9 = 119.9$
 $\therefore$ Sixth number = $63 + 68.4 - 119.9 = 11.5$
 Hence the middle number is = 11.5

33. (b); Total temp of mon + tues + wed + thus
 $= 4 \times 48^\circ\text{C} = 192^\circ\text{C}$
 Temp on mon = 42°C
 temp on Tue + Wed + Thus = $192^\circ - 42^\circ = 150^\circ\text{C}$
 Total temp for tue + wed + thus + Fri
 $= 52^\circ \times 4 = 208^\circ\text{C}$

Tem on Friday = $208^\circ - 150^\circ = 58^\circ\text{C}$

34. (a); Sum of first 25 multiples of 5 = $5(1 + 2 + 3 + \dots + 25)$

$$\frac{5 \times n(n+1)}{2} = 5 \times \frac{25 \times 26}{2} = 1625$$

$$\text{Average} = \frac{1625}{25} = 65$$

35. (c); Total expenditure of a man in year 1999
 $= 76535 + 88165 = 164700$

$$\text{Average} = \frac{164700}{365} = 451.23$$

36. (a); Let the 7th number be x.

According to the question

$$6 \times 8 + x = 7 \times 10 \Rightarrow x = 70 - 48 = 22$$

37. (c); Total expenditure for first 5 months = 120×5
 $= 600 \text{ Rs.}$

$$\text{Total expenditure for next 7 months} = 130 \times 7 = 910 \text{ Rs.}$$

Total income = Total expenditure of the whole year + saving

$$= 600 + 910 + 290 = 1800 \text{ Rs}$$

$$\text{Monthly average income} = \frac{1800}{12} = 150 \text{ Rs.}$$

38. (d); Total monthly salary of 20 workers in an office
 $= 1900 \times 20 = \text{Rs. } 38000$

Total monthly salary of 20 workers and manager
 $= 2000 \times 21 = \text{Rs. } 42000$

$$\text{Manager monthly salary} = 42000 - 38000 = \text{Rs. } 4000$$

$$\text{Manager annual salary} = 12 \times 4000 = \text{Rs. } 48000$$

39. (c); Total age of men = $15 \times 10 = 150$

$$\text{Total age of women} = 25 \times 12 = 300$$

$$\text{Average age of the whole club} = \frac{150 + 300}{40}$$

$$= \frac{450}{40} = 11.25 \text{ years.}$$

40. (a); Let the average of rest results = x

According to the question

$$14 \times 2 + x \times 10 = 15 \times 12 \Rightarrow 10x = 180 - 28$$

$$x = \frac{152}{10} \Rightarrow x = 15.2$$

Hence average of the rest is 15.2

Moderate

1. (a); Let the side of square be a
 and average speed of aeroplane x

$$\therefore \frac{a}{200} + \frac{a}{400} + \frac{a}{600} + \frac{a}{800} = \frac{4a}{x}$$

$$\Rightarrow \frac{12a + 6a + 4a + 3a}{2400} = \frac{4a}{x} \Rightarrow \frac{25}{2400} = \frac{4}{x}$$

$$\therefore x = 384$$

Hence average speed of a aeroplane = 384 km/hr

2. (d); Let the average speed of a cyclist = x

Total distance = 3 km

$$\therefore \frac{1}{20} + \frac{1}{16} + \frac{1}{12} = \frac{3}{x} \Rightarrow \frac{12 + 15 + 20}{240} = \frac{3}{x}$$

$$\frac{47}{240} = \frac{3}{x} \Rightarrow x = 15.32 \text{ km/hr}$$

Average speed of cyclist is 15.32 km/hr

3. (a); total marks obtained by 150 students.
 $= 100 \times 30 + 50 \times 60 = 6000$

$$\text{mean marks} = \frac{6000}{150} = 40$$

4. (b); Let the weight of teacher = x kg
 and average weight of students and teacher
 $= 42 + .40 = 42.40 \text{ kg}$
 $\therefore 34 \times 42 + x = 35(42.4) \Rightarrow x = 1484 - 1428 = 56 \text{ kg}$
 Hence weight of the teacher is 56 kg.

Shortcut:

The mean rises by 0.4 kg and total number of persons = 35

$$\text{Weight of teacher} = 42 + 0.4 \times 35 = 56 \text{ kg}$$

5. (a); Let the weight of teacher = x kg

According to the question

$$\therefore 21 \times 64 + x = 22 \times 65$$

$$x = 1430 - 1344 \Rightarrow x = 86 \text{ kg}$$

- Hence weight of the teacher is 86 kg.
Shortcut:
Teacher's weight = $64 + 22 \times 1 = 86$ kg
6. (a); Total price of all the articles.
 $= 5 \times 25 + 10 \times 50 + 15 \times 35$
 $= 125 + 500 + 525 = 1150$ Rs.
- Average price = $\frac{1150}{30} = 38.33$ Rs.
7. (a); Let the no of boys in a class = x
 $\therefore x \times 70 + (150 - x) 55 = 150 \times 60$
 $70x - 55x = 9000 - 8250 \Rightarrow 15x = 750$
 $x = 50 \Rightarrow$ No. of boys = 50
8. (d); Total score of a class of m students = $70 \times m$
Total score of a class of n students = $91 \times n$
 $\therefore 70m + 91n = (m + n) 80$
 $70m + 91n = 80m + 80n$
 $11n = 10m \Rightarrow \frac{n}{m} = \frac{10}{11}$
9. (b); Let the no of passed student = x
According to the question
 $x \times 39 + 15(120 - x) = 120 \times 35$
 $39x - 15x = 4200 - 1800$
 $24x = 2400 \Rightarrow x = 100$
Hence no of passes student is 100.
10. (a); Here a = 9 km, b = 25 km, c = 30 km
P = 3 km/hr Q = 5 km/hr R = 10 km/hr
 $\therefore$ Required average speed

$$= \frac{\frac{a}{P} + \frac{b}{Q} + \frac{c}{R}}{3} = \frac{\frac{9}{3} + \frac{25}{5} + \frac{30}{10}}{3} = \frac{9 + 5 + 3}{3} = \frac{17}{3} \text{ km/hr}$$

$$= \frac{64}{3+5+3} = \frac{64}{11} = 5\frac{9}{11} \text{ km/hr}$$

Hence average speed of a person = $5\frac{9}{11}$ km/h
11. (c); Total present age of Pand Q = $15 \times 2 + 10$
 $= 40$ years.
Total present age of P, Q and R
 $= 20 \times 3 = 60$ years.
P + Q + R = 60
(Present age) R = 20 years.
R age after 10 years = $20 + 10 = 30$ years.
12. (c); Let the total no of non officer staff in an office = x
According to the question
 $750 \times 15 + 250x = 500(x + 15)$
 $500x - 250x = 11250 - 7500$
 $250x = 3750 \Rightarrow x = \frac{3750}{250} = 15$
Hence no of non-officer staff = 15
13. (b); Total Present age of family of 5 members
 $= 27 \times 5 + 3 \times 5 = 135 + 15 = 150$ years
Total present age of a family including child
 $= 27 \times 6 = 162$ years
age of a child = $162 - 150 = 12$ years.
14. (d); Total age of 5 sisters = 100 years.
Total age of a group of sister at the birth of youngest sister = $100 - 4 \times 5 = 80$ years.
Average age of group of sisters $\frac{80}{4} = 20$ years.
15. (b); Sum of the weight of A, B, C and D = 67×4
 $= 268$ kg
Sum of the weight of A, B, C, D, E = $65 \times 5 = 325$ kg
weight of E = $325 - 268 = 57$ kg
Weight of F = $57 + 4 = 61$ kg
Total weight of F, B, C, D, E = $64 \times 5 = 320$ kg
Total weight of B, C, D, E = $320 - 61 = 259$ kg.
 $\therefore$ Weight of A = $325 - 259 = 66$ kg.
Shortcut:
Weight of E = $(67 - 2) - (4 \times 2) = 57$
Weight of F = 61 kg
Average weight decreases by 1 kg on replacing A.
So, weight of A = $61 + 5 \times 1 = 66$ kg
16. (a); Total of a + b + c = 33, c + d + e = 51
e + f = 44, e + c = 34, d = $51 - 34 = 17$
 $\therefore$ a + b + c + d + e + f = $33 + 17 + 44 = 94$
Average of a, b, c, d, e, f = $\frac{94}{6} = 15\frac{2}{3}$
17. (a); Total age of a man and his twin born son
 $= 30 \times 3 = 90$ years
given the ratio of father and his twin son
 $= 5 : 2 : 2$
Father's age = $\frac{90}{9} \times 5 = 50$ years
18. (c); Total age of Rakesh and Mohan = 30 years
Mohan + Ramesh = 24 years
Rakesh + Ramesh = 26 years
(Rakesh + Mohan + Ramesh) = $\frac{80}{2} = 40$ years.
age of Mohan = $40 - 26 = 14$ years.
19. (a); Let the second number = x
 $\frac{x}{2} + x + 2x = 84 \Rightarrow \frac{x + 2x + 4x}{2} = 84$

- $$7x = 84 \times 2 \Rightarrow x = \frac{84 \times 2}{7} \Rightarrow x = 24$$
- Third number = $24 \times 2 = 48$
20. (a); Total temp of Mon + tues + wed = 111°C
 tues + wed + thus = $34 \times 3 = 102^\circ\text{C}$
 Mon - Thus = 9°C
- $$M - \frac{4}{5}M = 9^\circ\text{C} \Rightarrow \frac{M}{5} = 9^\circ\text{C} \Rightarrow M = 45^\circ\text{C}$$
- Temp on thursday = $45 \times \frac{4}{5} = 36^\circ\text{C}$
21. (a); let the number be $(a - 5), (a - 3), (a - 1), (a + 1), (a + 3), (a + 5)$
 Then the average of all six consecutive odd numbers
- $$= \frac{(a - 5) + (a - 3) + (a - 1) + (a + 1) + (a + 3) + (a + 5)}{6} = a$$
- According to question,
- $$\frac{(a - 1)^2 + (a + 1)^2 + (a + 3)^2 + (a + 5)^2}{4} - \frac{(a - 5)^2 + (a - 3)^2 + (a - 1)^2 + (a + 1)^2}{4} = 64$$
- $$a = 8$$
22. (a); Let the number of candidates who passed = x
 According to question
 $55 \times x + 25(150 - x) = 150 \times 50$
- $$55x - 25x = 7500 - 3750 \Rightarrow x = 125$$
- Hence no of candidate who passed the exam = 125
23. (b); Total age of 24 boys and teacher = $(24 + 1) \times 15 = 375$ years
 Total age of all boys excluding teacher
 $= 24 \times (15 - 1) = 336$ years.
 age of teacher = $375 - 336 = 39$ years
Shortcut:
 Age of teacher = $15 + 24 \times 1 = 39$ years
24. (a); Total age of 8 members = $8 \times 40 = 320$ years.
 Total age after replacement of 1 member
 $= 320 - 55 + 39 = 304$ years.
- $$\text{New average} = \frac{304}{8} = 38 \text{ years}$$
- Hence average decreased by 2 years.
Shortcut:
 Difference in ages = $55 - 39 = 16$ years its effect
 $= \frac{16}{8} = 2$

25. (c); Sum of the age of 2 new players
- $$= 18 + 20 + 11 \times \frac{2}{12}$$
- $$= 38 + \frac{22}{12} = \frac{478}{12} \text{ years}$$
- average age of 2 new players
- $$= \frac{478}{12 \times 2} = \frac{478}{24} = 19\frac{22}{24}$$
- $$= 19 \text{ years } 11 \text{ months.}$$
26. (a); Let the number of students = x
 According to the question
 $x \times 6 + 12 \times 40 = (x + 12) \times 7$
 $6x + 480 = 7x + 84 \Rightarrow x = 396$
27. (a); Let the average of 11 innings = x
 Total run of 11 innings = $11x$
 After 12 innings
 $11x + 90 = (x + 5) \times 12$
 $12x - 11x = 90 + 60 \Rightarrow x = 150$
 Average after 12th innings = $150 - 5 = 145$
Shortcut:
 Average after 12 innings = $(90 + 12 \times 5) - 5 = 145$
28. (a); Let the no of wicket taken by the cricketer = x
 According to the question
- $$\frac{12.4 \times x + 26}{x + 5} = 12 \Rightarrow 12.4x + 26 = 12x + 60$$
- $$x = \frac{34}{.4} = \frac{340}{4} = 85$$
29. (c); Let his score in 26th inning = x
 According to the question
 $25 \times 56 + x = 26(56 + 2)$
 $x = 1308 - 1400 \Rightarrow x = 108$
Shortcut:
 Score in the 26th innings = $56 + 2 - 26 = 108$
30. (a); Let the age of a class teacher = x
 According to question
 $30 \times 13.5 + x = 14 \times 31 \Rightarrow x = 434 - 405 = 29$ years
Shortcut:
 Age of the teacher = $14 + 0.5 \times 30 = 29$ years
31. (a); Let the marks of New student = x
 $x = 90 - 20 \times 4 \Rightarrow x = 80$
 Marks of New student = 10

32. (a); Sum of Age of x and y = 40 years
y and z = 38 years, z and x = 42 years

$$x + y + z = \frac{40 + 38 + 42}{2} = \frac{120}{2} = 60$$

$$x = 60 - 38 = 22 \text{ years, } y = 60 - 42 = 18 \text{ years}$$

$$z = 60 - 40 = 20 \text{ years}$$

33. (a); Total age of a family of 6 members = 22×6
= 132 years

Total age of the family at birth of youngest member
= $132 - 7 \times 6 = 132 - 42 = 90$ years.

$$\text{Average age} = \frac{90}{5} = 18 \text{ years.}$$

34. (b); Age of a new man = $25 + 10 \times 2 = 45$ years.

35. (a); Total correct marks obtained by students
= $100 \times 60 - 75 + 65 = 5990$

$$\text{mean marks of 100 students} = \frac{5990}{100} = 59.9$$

Shortcut:

The actual marks are 10 less then the previous marks.

$$\text{So, correct mean} = 60 - \left(\frac{10}{100}\right) = 59.9$$

36. (c); Let the consecutives even numebr

$$= x, x + 2, x + 4, x + 6$$

$$\therefore x + x + 2 + x + 4 + x + 6 = 260$$

$$4x = 248 \Rightarrow x = 62$$

$$A \times D = 62 \times 68 = 4216$$

37. (c); Total correct weight of 36 students
= $50 \times 36 - 73 + 3 = 1800 - 36 = 1764$ kg.

$$\text{Correct average} = \frac{1764}{36} = 49 \text{ kg.}$$

Shortcut:

$$\text{Correct average} = 50 - \left(\frac{73 - 37}{36}\right) = 49 \text{ kg}$$

38. (c); Sachin score in 26th innings.
= $26 \times (46 + 2) - 25 \times 46 = 1248 - 1150 = 98$

Shortcut:

$$\text{Sachin's score in 26th innings} = 46 + 2 \times 26 = 98$$

39. (a); total present age of 5 members = $17 \times 5 + 3 \times 5$
= 100

$$\text{total present age including new baby} = 17 \times 6$$

$$= 102$$

$$\text{Age of new baby} = 102 - 100 = 2 \text{ years.}$$

40. (b); Total sum of all the 5 subjects = $75 \times 4 + 80$
= 380

$$\text{Average} = \frac{380}{5} = 76$$

Shortcut:

$$\text{New average} = 75 + \left(\frac{80 - 75}{5}\right) = 76 \text{ marks}$$

Difficult

1. (b); Total weight of A + B + C = $84 \times 3 = 252$ kgs

$$A + B + C + D = 80 \times 4 = 320 \text{ kgs.}$$

$$D = 68, E = 68 + 3 = 71 \text{ kgs}$$

$$B + C + D + E = 79 \times 4 = 316 \text{ kg}$$

$$B + C + D = 316 - 71 = 245 \text{ kg}$$

$$A = 320 - 245 = 75 \text{ kgs.}$$

2. (a); Total runs of 42 innings = $42 \times 30 = 1260$

Total run excluding min and max inning

$$= 40 \times 28 = 1120$$

$$\text{max} + \text{min} = 1260 - 1120 = 140 \quad \dots(i)$$

$$\text{max} - \text{min} = 100 \quad \dots(ii)$$

On solving (i) & (ii)

$$\text{max} = 120, \text{ min} = 20$$

Hence, minimum score is 20.

3. (b); Sum of the present age of Husband and wife

$$= 23 \times 2 + 5 \times 2 = 56 \text{ years}$$

Sum of the present age of Husband, wife and child

$$= 20 \times 3 = 60$$

$$\text{Age of child} = 60 - 56 = 4 \text{ years.}$$

4. (b); Present age of Husband and wife = $25 \times 2 + 7 \times 2$
= 64 years

Present age of Husband wife and child

$$= 22 \times 3 = 66 \text{ years}$$

$$\text{Age of a child is} = 66 - 64 = 2 \text{ years.}$$

5. (a); Sum of the present age of Mother, Father and Son
= $42 \times 3 + 6 \times 3 = 126 + 18 = 144$ years.

$$\text{Sum of present age of the family} = 36 \times 5 = 180$$

$$\text{Present age of the bride} = 180 - 144 - 5 = 31 \text{ years.}$$

$$\text{Age of the bride at the time of marriage}$$

$$= 31 - 6 = 25 \text{ years.}$$

6. (a); Let the average expenditure of a man = x Rs.

$\therefore$ According to the question

- $\therefore 35 \times x + 42 = 42(x - 1)$
 $\therefore 7x = 84 \Rightarrow x = 12 \text{ Rs.}$
 Original expenditure = $35 \times 12 = 420 \text{ Rs.}$
- 7. (d);** Ashish total earnings of 11 months
 $= 4200 \times 11 = 46,200\$$
 Total earnings to be needed for 12 months
 $= 5000 \times 12 = 60,000\$$
 Last month earnings = $60,000 - 46,200 = 13,800\$$
Shortcut:
 Required earning in the last month
 $= 5000 + (5000 - 4200) \times 11 = \$ 13800$
- 8. (a);** Let the three no is x, y, z
 $x : y \Rightarrow (2 : 3) \times 5 \Rightarrow y : z = (5 : 8) \times 3$
 $\Rightarrow x : y : z = 10 : 15 : 24$
 Second number $y = \frac{15}{49} \times 98 = 30$
- 9. (a);** Let the total number of workers = x
 $\therefore 15 \times 525 + (x - 15) 85 = x \times 95$
 $\therefore 15 \times 525 - 15 \times 85 = 10x$
 $\therefore 7875 - 1275 = 10x \Rightarrow 10x = 6600 \Rightarrow x = 660$
- 10. (d);** Let his Avg be A
 And increased average = $A + x$
 $86A + 270 = 87(A + x) \Rightarrow x = \frac{(270 - A)}{87}$
 if x be a whole number possible value of A
 $A = 9, 96, 183, 270$ for $x = 3, 2, 1, 0$
 So new value of $A = 12, 98, 184, 270$
- 11. (b);** Let the Average expenditure of the party is x
 $\therefore 13 \times 79 + 6(x + 4) = 19 \times x$
 $13x = 13 \times 79 + 24 \Rightarrow x = \frac{1027 + 24}{13} = 80.84$
 Total expenditure of the party = $80.84 \times 19 = 1536.07$
- 12. (b);** Let the person initially going for picnic is x
 $\therefore 16.75 \times x + 20 \times 13.25 = 15(x + 20)$
 $1.75x = 15 \times 20 - 20 \times 13.25$
 $1.75x = 35 \Rightarrow x = 20$
- 13. (c);** Let the total no of faculty = x
 and management trainees = y
 $4000x + 1250y = (x + y)2000$
 $8x = 3y \Rightarrow x : y = 3 : 8$
 So the possible value could be = 110
- 14 (a);** 10 years ago average age of 25 teachers = 45 years.
 4 years ago (just before the retirement of principal)
 average age of 25 teachers = $45 + 6 = 51$
 and at the same time total age of 25 teachers
 $= 51 \times 25 = 1275$
 and the total age of remaining 24 teachers when
 just principal has retired = $1275 - 60 = 1215$ years
 1 year later (i.e. 3 years ago from present)
 total age of 24 teacher (just before the recruitment
 of new principal) = $1215 + (1 \times 24) = 1239$ years.
 and the total age of 25 teacher including new
 principal just after the recruitment = $1239 + 54$
 $= 1293$ years.
 Thus the present age of all the 25 teachers
 $= 1293 + 3 \times 25 = 1368$ years.
 Hence the present average age of the 25 teacher.
 $= \frac{1368}{25} = 54 \frac{18}{25}$ years.
- 15. (a);** Sum of six lightest weight players = 49×6
 $= 294 \text{ kgs.}$
 sum of six heaviest players weight = 52×6
 $= 312 \text{ kg}$
 $\therefore$ weight of the player at 6th position
 $= 312 + 294 - 550 = 56 \text{ kg.}$
- 16. (a);** Total age of $S + G = 35 \times 2 = 70$ years
 $K + G = 32 \times 2 = 64$ years
 $S + K = 38 \times 2 = 76$ years
 $S + G + K = \frac{70 + 64 + 76}{2} = \frac{210}{2} = 105$ years
 Average of $S + G + K = \frac{105}{3} = 35$ years
 $D + I = \frac{35}{2} \Rightarrow$ Sum of age of $D + I = 35$
 Average of $S, G, K, D, I = \frac{105 + 35}{5} = 28$ years
- 17. (c);** Total present age of a family of 6 members
 $= 25 \times 6 + 4 \times 6 = 174$
 Total present age of a family including child = $25 \times 7 = 175$ years.
 Age of child is = $175 - 174 = 1$ year.
- 18. (a);** Total expenditure of Jan to June = 4200×6
 $= \text{Rs. } 25200$
 Total expenditure from Feb to July
 $= 25200 - 1200 + 1500 = \text{Rs. } 25500$
 Average = $\frac{25500}{6} = \text{Rs. } 4250$
- 19. (a);** The price of the costliest and cheapest computers
 $= (80 \times 30,000) - (78 \times 29,500) = 99,000$
 Therefore the price of cheapest computer
 $= 99,000 - 80,000 = 19,000$

20. (a);

	No. of family member	Average	Total
Eleven years Earlier	4	28	112
Presently	if 4	39	156
	6	28	168

Since it is obvious that the youngest member (i.e. child) was the 6th family member in the family. Therefore at the time of the birth of the youngest child the elder child's age was 6 years.

Now the sum of their ages

Let age of youngest member = x

$$x + x + 6 = (168 - 156) \Rightarrow 2x + 6 = 12 \Rightarrow x = 3$$

Then the present age of the youngest member of the family is 3 years.

21. (c); let the average of walking = x km/hr

$$\text{Total distance} = 32 \times 3 = 96$$

$$\therefore \frac{6}{x} + \frac{90}{60} = 3, \therefore \frac{6}{x} + \frac{3}{2} = 3$$

$$\frac{12 + 3x}{2x} = 3 \Rightarrow 3x = 12 \Rightarrow x = 4 \text{ km/hr.}$$

22. (a); Let the average of prime number be P and average of composite number be C.

Again the number of prime number be x, then the number of composite number be 2x

$$\frac{Px + 2Cx}{3x} = 9 \Rightarrow P + 2C = 27$$

$$\frac{2Px + Cx}{3x} = 11 \Rightarrow 2P + C = 33$$

On adding eq. (i) and (ii)

$$\text{we get } P + C = 20$$

and on Subtracting eq. (i) from (ii)

$$\text{we get } P - C = 6$$

$$\text{Therefore } P = 13 \text{ and } C = 7 \Rightarrow \text{thus } \frac{C}{P} = \frac{7}{13}$$

23. (c); Let the number of subjects be n and average marks be x, the total marks = nx

$$\text{Again } (n + 1)(x - 1) = (nx - 40) + (23 + 25)$$

$$x - n = 9 \quad \dots(i)$$

$$\text{Further } (n + 2)(x + 1) = (nx - 40) + (23 + 25) + 57$$

$$nx + 2x + n + 2 = nx + 65$$

$$2x + n = 63 \quad \dots(ii)$$

Solving equation I and II we get

$$n = 15 \text{ \& } x = 24$$

24. (a); Let Donald be denoted by H (Husband)

His wife be denoted by W, His daughter be denoted by D, His son be denoted by S

The average age of 4 persons

$$= \frac{H + W + D + S}{4} = 23$$

$$H + W + D + S = 92 \Rightarrow H = W + 4$$

	H	W	D	S
At the time when daughter is born	28 $\xleftarrow{(+4)}$	24	0	x
	↓			
At the time when Son is born	32 $\xleftarrow{(-4)}$	28	4	0

So at the time of birth of his Son, total age of his family = 64 years. (32 + 28 + 4 + 0 = 64) and presently the total age of his family = 92 years. It means total increase in age of the whole family = 28 years.

$$\text{Thus average increase in age} = \frac{28}{4} = 7 \text{ years.}$$

It means the age of Donald = 39 years

and age of his daughter = 11 years

Therefore the average age of Donald and his Daughter is 25 years.

25. (b); let the number be (a - 5), (a - 3), (a - 1), (a + 1), (a + 3), (a + 5)

Then this average of all six consecutive odd numbers

$$= \frac{(a - 5) + (a - 3) + (a - 1) + (a + 1) + (a + 3) + (a + 5)}{6} = a$$

Again the value of 'a' can be found by using the last statement

$$(a - 5)^2 + (a + 5)^2 = 178 \Rightarrow a^2 = 64 \Rightarrow a = 8$$

Average of all six number is 8.

26. (a);

	No of Directors	Average Age	Total age
Just before death and resignation	10	48	480
Just after death and resignation	9		(480 - (53 + x) + 34)
one year later	9	46	414

So one year later, after the incident

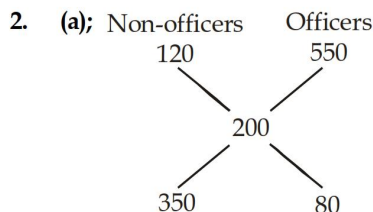
$$\text{total age} = \{480 - (53 + x) + 34\} + 9 \times 1 = 414$$

$$x = 56$$

where x is the age of the dead person at the time of his death.

Previous Year (Memory Based)

1. (a); Total age of 11 player and coach = 12×18
 = 216 yr.
 Total age of 11 player = $11 \times 17 = 187$ yr
 Age of coach = $216 - 187 = 29$ yrs.



Ratio = 35 : 8

No. of officers $8x \rightarrow 16, \quad x \rightarrow 2$

So, no. of non-officers = $35 \times 2 = 70$

3. (a); Required Avg. speed = $\frac{D_1 + D_2 + D_3}{T_1 + T_2 + T_3}$
- $$= \frac{20 + 15 + 10}{\frac{20}{5} + \frac{15}{3} + \frac{10}{2}} = \frac{45}{14} = 3\frac{3}{14} \text{ km/h}$$

4. (b); Req. Avg. speed = $\frac{1}{P_1 + P_2 + P_3}$
- $$P_1 = \frac{1}{4 \times 10}, \quad P_2 = \frac{9}{20 \times 5}, \quad P_3 = \frac{3}{10 \times 15}$$
- So, $= \frac{1}{\frac{1}{4 \times 10} + \frac{9}{20 \times 5} + \frac{3}{10 \times 15}} = 7.40 \text{ km/h}$

5. (b); Let min. score = x
 Max. score = $100 + x$
 So, $x + (x + 100) = 30 \times 42 - 40 \times 28$
 $x = 20$
 Thus, the max. score = $x + 100 = 120$

6. (b); Total Average = $\frac{20 \times 80 + 30 \times 70}{20 + 30} = 74\%$

7. (d); Let the total no. of student be 100
 Also, let the mean score = x
 According to the question
 $20 \times 80 + 25 \times 31 + (100 - 20 - 25)x = 100 \times 52$
 $x = 51.36 = 51.4\%$

8. (c); No. of orange bought by the person
- $$= \frac{36}{1} + \frac{36}{1.50} + \frac{36}{1.80} + \frac{36}{2} + \frac{36}{2.25} = 114$$
- Total expenditure = $36 \times 5 = 180$
- Avg. of each orange = $\frac{180}{114} = 1.58$

9. (b); Let no. of students in section A, B, C and D be w, x, y, z .
 So, Average weight of students of section A and B = 48 kg

$$\frac{45w + 50x}{w + x} = 48$$

$$15w = 10x$$

...(i)

Avg. weight of students of sections B and C = 60kg.

$$50x + 72y = 60(x + y)$$

$$10x = 12y$$

...(ii)

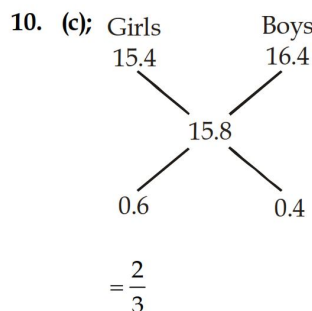
Avg. weight of student A, B, C and D is 60 kg.

$$45w + 50x + 72y + 80z = 60(w + x + y + z)$$

$$15w + 10x - 12y - 20z = 0$$

$$15w = 20z$$

$$\frac{w}{z} = \frac{4}{3}$$



11. (a); Required Average = $\frac{2 + 3 + 5 + 7 + 11}{5} = 5.6$

12. (b); When we calculating the Average of run in cricket matches, not out innings are not considered.

So, Total runs

$$= 53 + 52 + 11 + 0 + 5 + 104 + 35 = 260$$

$$\text{Average of the player inserees} = \frac{260}{5} = 52 \text{ runs}$$

13. (c); Let the no. of children in the family is x .

$$(5 + 4 + x) \times \frac{3}{2} = 5 \times 1 + 4 \times \frac{5}{4} + x \times 2$$

$$(9 + x) \times 3 = 2(5 + 5 + 2x)$$

$$x = 7$$

14. (d); Temp. on thursday = Temp. on Monday + Avg. increase in the temp. of three days.
 = $29^\circ + 3 \times 2^\circ = 35^\circ$

15. (b); New Average = $28 + \frac{(24 - 42) + (62 - 12)}{40}$

$$= 28 + \frac{32}{40} = 28.8$$

16. (c); Required number = $15 \times 27 - 7 \times 25 - 7 \times 31 = 13$

17. (a); Sum of ages of 9 boys = $15 \times 9 = 135$ years
Sum of ages of 9 boys + 1 teacher = $17 \times 10 = 170$ years

The age of teacher = $170 - 135 = 35$ years

18. (c); New Average

$$= \text{Avg. height} + \frac{\left(\begin{array}{l} \text{Change of avg. of going} \\ \text{and coming student} \end{array} \right) \times 10}{50}$$

$$= 152 + \frac{(150 - 148) \times 10}{50} = 152.4 \text{ cm}$$

19. (a); Average of all 8 boys = $\frac{5 \times 12 + 3 \times 16}{8}$

$$= 13\frac{1}{2} \text{ years}$$

20. (c); Average of the entire group

$$= \frac{40 \times 4200 + 35 \times 4000}{75} = 4106\frac{2}{3}$$

21. (c); Average production of last 5 days

$$\Rightarrow \frac{60 \times 25 + 5 \times x}{30} = 58 \text{ is } x,$$

$$\text{So, } x = \frac{30 \times 58 - 60 \times 25}{5} = 48$$

22. (d); Let the no. of another labour = x

The no. of agriculture labour = $11x$

Total annual income = $11x \times S + x \times T$

$$\text{Average annual income} = \frac{(11S + T)x}{12x} = \frac{11S + T}{12}$$

$$\text{Average monthly income} = \frac{11S + T}{12 \times 12} = \frac{11S + T}{144}$$

23. (c); $8 \times 15 + x \times 6 = (8 + x) \times 10.8$
 $x = 7$

24. (b); A.T.Q.

$$7 \times 12000 + x \times 6000 = (7 + x) \times 8000$$

$$x = 14$$

Total workers = $14 + 7 = 21$

25. (a); A.T.Q.

$$x \times 39 + (120 - x) \times 15 = 120 \times 35$$

$$x = 100$$

26. (b); Correct average wages

$$= 200 + \frac{80 + 220 - 180 - 20}{500} = 200.20$$

27. (a); $8 \times 3160 + 5 \times 4120 - 3400 \times 12 = 5080$

28. (c); Let the player took x wickets at the average of 24.85
then $24.85x + 52 = 24(x + 5)$

$$x = 80$$

total wickets = $80 + 5 = 85$

29. (c); A.T.Q.

$$\Rightarrow \frac{19x + 100}{20} = (x + 2), \quad x = 60$$

So, Avg. runs in 20 innings = $60 + 2 = 62$

30. (b); Avg. rainfall

$$= \frac{20.54 + 33.10 + 11.62 + 19.20 + 21.74}{5}$$

$$= 21.24 \text{ inch}$$

31. (d); $\frac{x + 5 + x + 7}{2} - \frac{x + x + 2 + x + 4}{3} = 4$

$$\text{So, } \frac{(2x + 12)}{2} - \frac{(3x + 6)}{3} = 4, \quad 4 = 4$$

This is correct for any no. so, cannot be determined. Since x is cancelled out

32. (b); The average for set $A = \frac{376}{8} = 47$

The smallest no. of second set = $47 + 15 = 62$

Required total = $62 + 63 + 64 + 65 + 66$

$$= 64 \times 5 = 320$$

33. (d); Let initial expenditure on food per day = x

$$\frac{x + 42}{35 + 7} = \frac{x}{35} - 1$$

$$\Rightarrow 35x + 1470 = 42x - 1470, \quad x = 420$$

34. (d); The total of first 5 natural no. = 15

the total of first 8 natural no. = 36

$$\text{Increase in Average} = \frac{36}{8} - \frac{15}{5} = 1.5$$

35. (c); The total of m numbers = $n \cdot n^2$

and the total of n numbers = $n \cdot n^2$

Required Avg. = $(m + n)$ avg. of the numbers.

$$= \frac{mn^2 + nm^2}{(m + n)} = mn$$

36. (b); Weight of new student = Original value + no. $\times$ increase

$$= 51 + 50 \times \frac{1}{2} = 76 \text{ kg.}$$

37. (d); Average marks of all 4 sets = $\frac{3N + N + 20}{4}$

$$= N + 5$$

38. (a); The total difference of the total of the no.

$$\Rightarrow (10x + y) - (10y + x) = 1.8 \times 10, \quad x - y = 2$$

39. (b); The Average of first and the last no. will be same as the average of second no. and the second last number. so, Average is = 6.5

40. (b); New Average is = $13 + \frac{73 - 48}{25} = 14$